Text-Based Invoice Data Extraction

1. Problem Statement

Develop a Python program to process invoice strings, extract product codes, quantities, and amounts, and identify invalid invoice records.

2. Objectives

- Extract product codes.
- Extract quantities and amounts.
- Process invoice strings.
- Identify malformed records.
- Calculate total invoice amount.

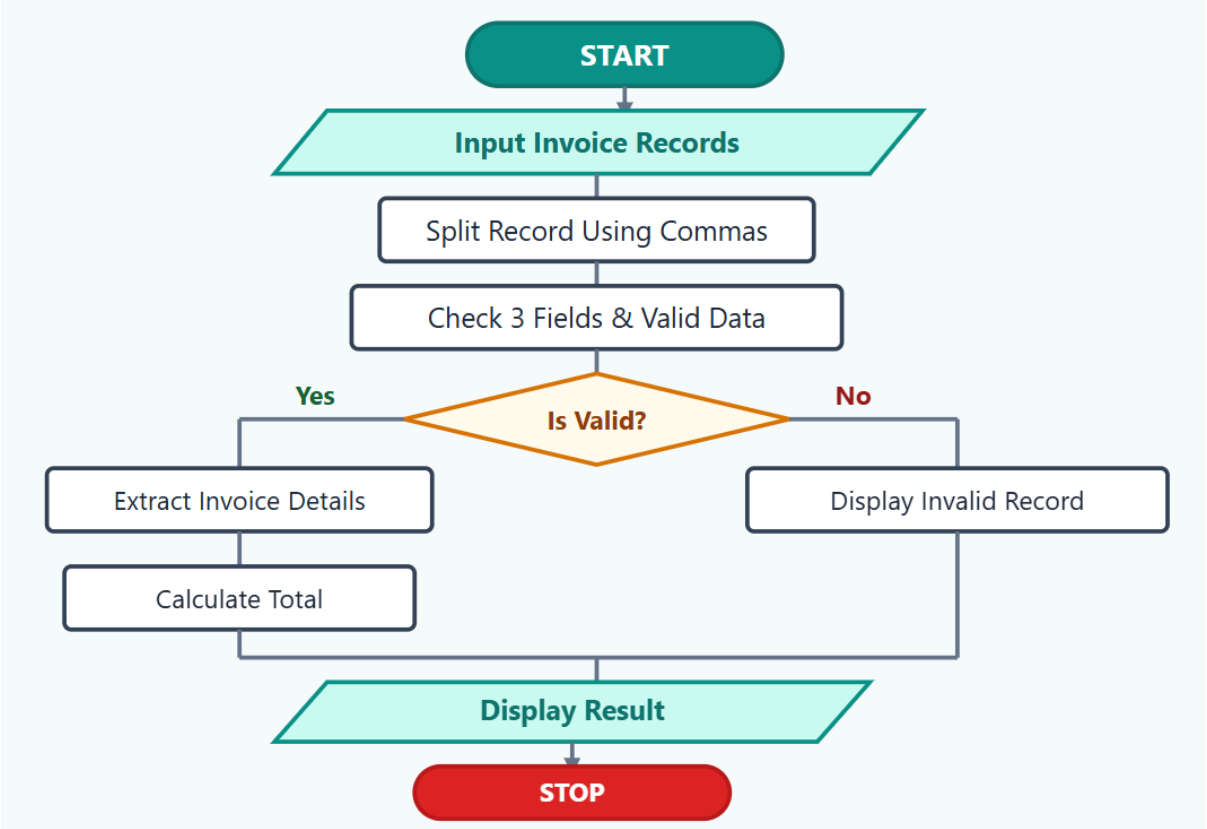
3. Assumptions

- Invoice format: `CODE,QUANTITY,AMOUNT`
- Quantity must be a positive integer.
- Amount must be positive.
- Each record contains exactly three fields.

4. Algorithm

1. Start.
2. Store invoice records.
3. Split each record using commas.
4. Extract product code, quantity, and amount.
5. Validate the record.
6. If valid, calculate total amount.
7. Otherwise, display an error.
8. Display invoice details and total.
9. Stop.

5. Flowchart



None

6. Pseudocode

None

```
START

Store invoice records
total = 0

FOR each record:
    Split record using comma

    IF record has 3 fields:
        Extract code, quantity, amount

        IF quantity and amount are valid:
```

```
            Display invoice details
            Add amount to total
        ELSE:
            Display Invalid Record
    ELSE:
        Display Malformed Record

Display total amount

STOP
```

8. Sample Output

```
None
P101 2 500.0
P102 3 750.0
Invalid Record: P103,abc,400
P104 1 300.0

Total Amount: 1550.0
```

9. Test Cases

Test	Input	Expected Result
1	P101,2,500	Valid
2	P102,abc,400	Invalid
3	P103,0,200	Invalid

4	P104,1,300	Valid
5	P105,2	Malformed

10. Conclusion

The program extracts invoice details using Python string operations and identifies malformed records. It also calculates the total invoice amount.